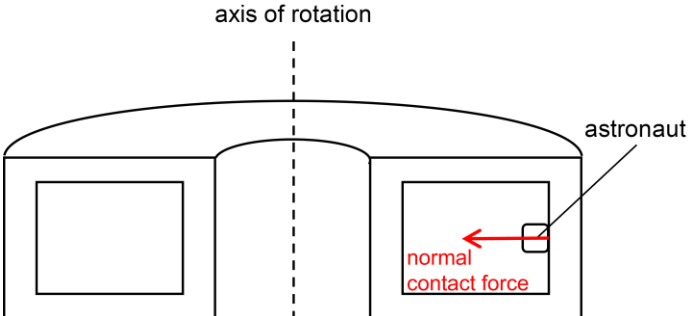


Qn	Suggested Solutions
3(a)	
(b)(i)	Although the magnitude of the velocity is constant, its velocity is changing as the direction of its velocity is always changing.
	According to Newton's Second Law, the astronaut must experience rate of change of momentum / an acceleration and hence a resultant force.
	Both the acceleration and force act towards the centre.
(ii)	$g = 9.81 = \frac{v^2}{r}$ $r = \frac{100^2}{9.81} = 1019$
	$r = 1020m$

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